

File permissions in Linux

Project description

In this project, I learned how Linux file permissions work and how to manage access using basic commands. I used Linux commands to check file permissions, interpret what they mean, and make changes to improve security. The goal was to make sure files and directories only had the correct level of access for users, groups, and others.

Check file and directory details

I used the following command to view all files, including hidden ones, and see their permissions: `ls -la /home/researcher2/projects`. This command showed me a detailed list of files and directories, including permission settings, ownership, and hidden files like `.project_x.txt`.

Describe the permissions string

An example of a permissions string is: `-rw-r-----`. This 10-character string shows file type and access levels. The first character shows if it is a file or directory (`-` means file, `d` means directory). The next three characters are for the user (owner), then group, then others. In this example, the user can read and write, the group can only read, and others have no access. This helps control who can access or modify a file.

Change file permissions

To fix incorrect permissions where others had write access, I used the following command: `chmod o-w /home/researcher2/projects/project_k.txt`. This removed write permission from “others” so only the correct users can modify or view the file based on the organization’s security rules. I then verified the change using `ls -l`.

Change file permissions on a hidden file

The hidden file `.project_x.txt` needed stricter permissions so that it could not be written to by anyone except the correct users. I used: `chmod 640 /home/researcher2/projects/.project_x.txt`. This gave the owner read/write access, the group read access, and removed all permissions for others. This ensures the hidden file is protected but still readable by the right people.

Change directory permissions

To secure the `drafts` directory so only the owner can access it, I used: `chmod 700 /home/researcher2/projects/drafts`. This removed access from group and others and left full access only for the owner (`researcher2`). This is important for protecting sensitive or unfinished work inside the directory.

Summary

In this project, I practiced using Linux commands to manage file and directory permissions. I learned how to use `ls -la` to check permissions and `chmod` to change them based on security requirements. I also worked with hidden files and restricted directory access to improve overall system security. This helped me understand how Linux controls access and how important proper permissions are in cybersecurity.